Forces And Friction

Solutions

Overview

Resolving forces

Replace a force by perpendicular components. Use these to find a resultant or apply Newton's second law.

Inclined planes

Choose axes parallel and perpendicular to the plane.

Friction

Distinguish friction below its limiting value from limiting friction.

Unless stated otherwise, use $g = 9.8\,\mathrm{m\,s^{-2}}$.

Resolving Forces

Solution 3

A displacement of 5 m makes 60° above the horizontal. Find its horizontal and vertical components.

Displacement

$x = 5\cos 60^\circ = 2.5\,\text{m},$

$y = 5\sin 60^\circ = 4.33\,\text{m}.$

A force of 7 N makes 60° to the upward vertical, towards the right. Resolve it.

Force

$F_x = 7\sin 60^\circ = 6.06\,\text{N},$

$F_y = 7\cos 60^\circ = 3.5\,\text{N}.$

Tip: the component adjacent to the stated angle uses cosine; the opposite component uses sine.

Quickfire Questions

Solution 4

Resolve 8 N, 30° above the positive horizontal.

a

$4\sqrt{3}\mathbf{i} + 4\mathbf{j}\ \text{N}.$

Resolve 6 N, 45° above the negative horizontal.

b

$-3\sqrt{2}\mathbf{i} + 3\sqrt{2}\mathbf{j}\ \text{N}.$

Resolve 9 N, 37° below the positive horizontal.

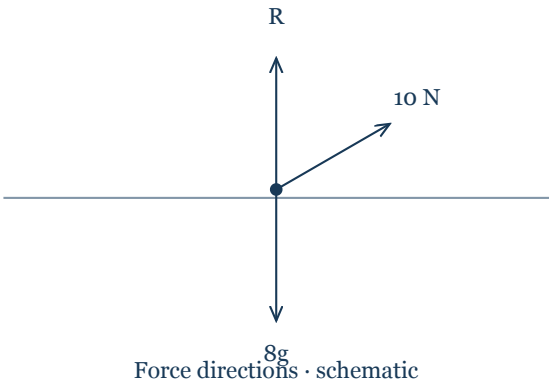
c

$7.19\mathbf{i} - 5.42\mathbf{j}\ \text{N}.$

Applied Example

Solution 5

A box of mass 8 kg rests on a smooth horizontal floor. A force of 10 N pulls at 30° above the horizontal. Find the acceleration and normal reaction.



Force directions · schematic

Answer

Horizontal:

$$8a = 10\cos 30^\circ \Rightarrow a = \frac{5\sqrt{3}}{8}\ \text{m s}^{-2}.$$

Vertical equilibrium:

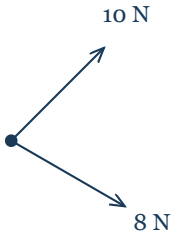
$$R + 10\sin 30^\circ = 8g$$

$$\Rightarrow R = 8g - 5 = 73.4\ \text{N}.$$

Combining Forces

Solution 6

Find the resultant of 10 N at 45° above the positive horizontal and 8 N at 30° below it.



Force directions · schematic

Answer

$$F_x = 10\cos 45^\circ + 8\cos 30^\circ = 13.999,$$

$$F_y = 10\sin 45^\circ - 8\sin 30^\circ = 3.071.$$

$$|\mathbf{F}| = \sqrt{F_x^2 + F_y^2} = 14.3\ \text{N}.$$

$$\theta = \tan^{-1}(F_y/F_x) = 12.4^\circ$$

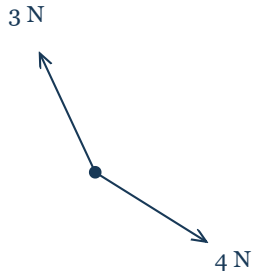
above the positive horizontal.

Alternatively, use the cosine rule with the 105° internal angle of the force triangle, then the sine rule.

Test Your Understanding

Solution 7

Find the resultant of 3 N, 25° left of the upward vertical, and 4 N, 32° below the positive horizontal.



Force directions · schematic

Answer

$F_x = 4\cos 32^\circ - 3\sin 25^\circ = 2.124,$

$F_y = 3\cos 25^\circ - 4\sin 32^\circ = 0.599.$

$|\mathbf{F}| = 2.21\,\text{N}, \quad \theta = 15.8^\circ$

above the positive horizontal.

Exercise 5A – Y2 Mech Pages 94-96

Solution 8

Exercise 5A · Question 4

Three forces are in equilibrium. Angles below are measured anticlockwise from the positive horizontal. Find the magnitude and direction of B.

(a) 15 N at 150°; 20 N at 210°.

a

$B_x = 35\cos 30^\circ, \quad B_y = 5\sin 30^\circ.$

$B = 30.4\,\text{N}, \quad \theta = 4.72^\circ$

above the positive horizontal.

(b) 25 N at 50°; 10 N at −30°.

b

$B_x = -25\cos 50^\circ - 10\cos 30^\circ,$

$B_y = -25\sin 50^\circ + 10\sin 30^\circ.$

$B = 28.5\,\text{N}, \quad \theta = 29.8^\circ$

below the negative horizontal.

(c) 10 N at 120°; 20 N at −20°.

c

$B_x = 5 - 20\cos 20^\circ,$

$B_y = -10\sin 60^\circ + 20\sin 20^\circ.$

$B = 13.9\,\text{N}, \quad \theta = 7.52^\circ$

below the negative horizontal.

Exercise 5A · Question 6

A 10 kg particle on a smooth horizontal surface is pulled by P at 45° above the horizontal. Its acceleration is 2 m s^{-2}. Find P. (3 marks)

6

$P\cos 45^\circ = 10(2)$

$\Rightarrow P = 20\sqrt{2}\,\text{N}.$

Question 7

A particle of mass m on a smooth horizontal surface is pulled by 20 N at 25° above the horizontal. Its acceleration is 0.5 m s^{-2}. Find m. (3 marks)

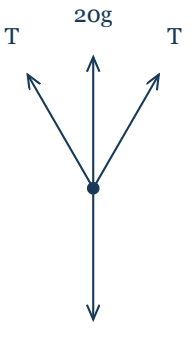
7

$20\cos 25^\circ = 0.5m$

$\Rightarrow m = 40\cos 25^\circ = 36.3\,\text{kg}.$

Exercise 5A · Question 8

A parachutist of mass 80 kg descends at constant velocity. Two supporting lines each make 60° with the horizontal and have tension T. Air resistance on the parachutist is one quarter of the weight. Find T.



Force directions · schematic

Answer

Air resistance is $20g$ N upwards.

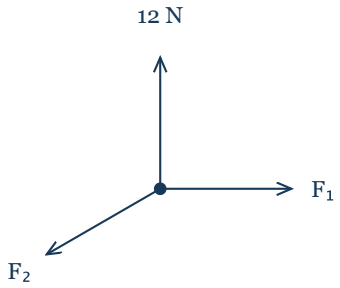
Vertical equilibrium gives

$2T\sin 60^\circ + 20g = 80g.$

$T = \dfrac{60g}{2\sin 60^\circ} = 20\sqrt{3}g\,\text{N}.$

Exercise 5A · Question 9

Forces act at a point: 12 N vertically upwards, F_1 horizontally right, and F_2 at 30° below the negative horizontal. Their resultant is $(2\sqrt{3}\mathbf{i} + 2\mathbf{j})$ N. Find F_1, F_2.



Force directions · schematic

Answer

Vertical components:

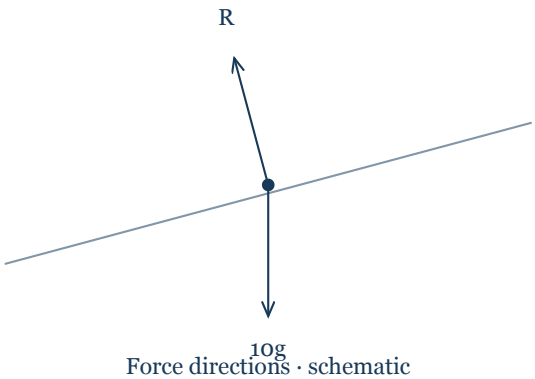
$12 - F_2 \sin 30^\circ = 2 \Rightarrow F_2 = 20\,\text{N}.$

Horizontal components:

$F_1 - 20\cos 30^\circ = 2\sqrt{3}$

$\Rightarrow F_1 = 12\sqrt{3}\,\text{N}.$

A particle of mass 10 kg slides down a smooth plane inclined at 15°. Find the normal reaction and acceleration.



Force directions · schematic

Answer

Choose axes parallel and perpendicular to the plane.

Perpendicular equilibrium:

$R = 10g\cos 15^\circ = 94.7\,\text{N}.$

Down the plane:

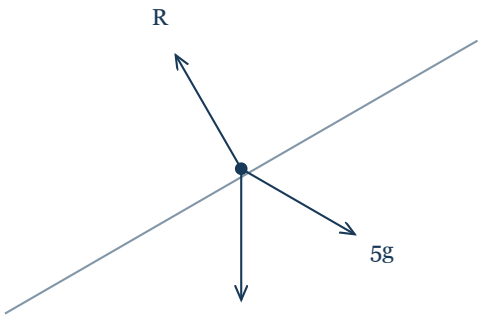
$10g\sin 15^\circ = 10a$

$\Rightarrow a = g\sin 15^\circ = 2.54\,\text{m s}^{-2}.$

Inclined Plane with an additional force

Solution 13

A particle of mass m is pushed up a smooth 30° slope by a force of $5g$ N. The force makes 60° with the upward slope, directed into the plane. The acceleration is $0.5\,\mathrm{m\,s^{-2}}$ up the slope. Find m.



Force directions · schematic

Answer

Resolve up the slope:

$$5g\cos 60^\circ - mg\sin 30^\circ = 0.5m.$$

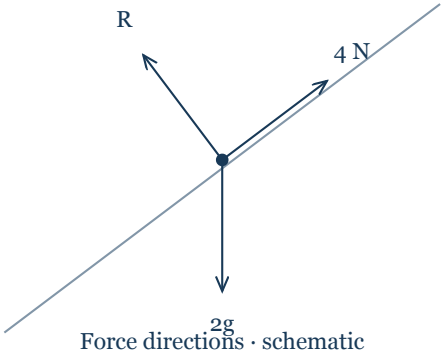
$$5g - mg = m$$

$$\Rightarrow m = \frac{5g}{1+g} = 4.54\,\mathrm{kg}.$$

Test Your Understanding

Solution 14

A 2 kg particle on a smooth plane is pulled by 4 N up the plane. The inclination α satisfies $\tan\alpha = 3/4$. Find the acceleration.



Force directions · schematic

Answer

Using a 3, 4, 5 triangle, $\sin\alpha = 3/5$.

Taking up the slope as positive:

$$4 - 2g\frac{3}{5} = 2a \Rightarrow a = -3.88.$$

The acceleration is $3.88\,\mathrm{m\,s^{-2}}$ down the slope.

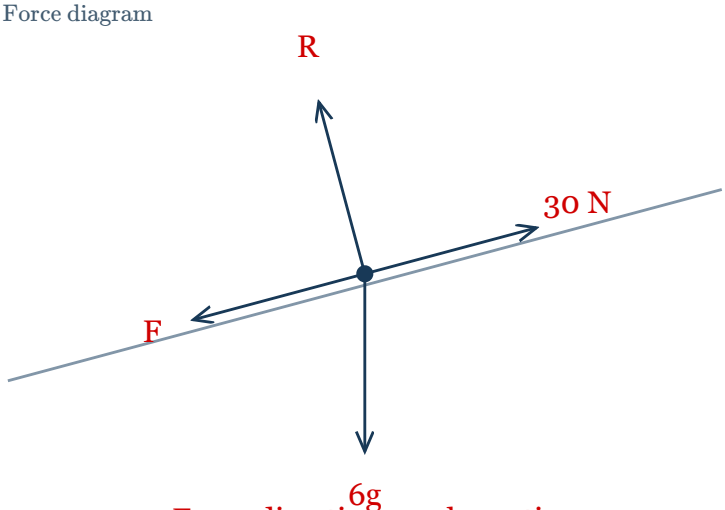
Exercise 5B · Question 4

A 6 kg particle is pulled up a rough 15° slope by a 30 N force parallel to the plane, at constant velocity.

(a) Draw a force diagram. (4 marks)

(b) Find the frictional force. (5 marks)

Force diagram



Force directions · schematic

Answer

(a) Weight $6g$ vertically down; reaction perpendicular out of the plane; 30 N up the plane; friction F down the plane.

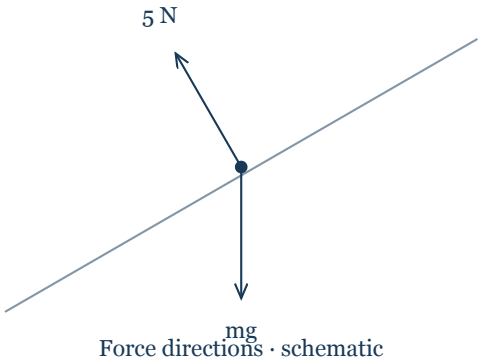
(b) Up the plane:

$$30 - 6g\sin 15^\circ - F = 0$$

$$F = 30 - 6g\sin 15^\circ = 14.8\,\text{N}.$$

Exercise 5B · Question 5

A particle of mass m slides down a smooth 30° plane. The normal reaction is 5 N. Find (a) m (3 marks); (b) its acceleration (3 marks).



Force directions · schematic

Answer

(a)

$$mg\cos 30^\circ = 5$$

$$m = \frac{5}{g\cos 30^\circ} = 0.589\,\text{kg}.$$

(b)

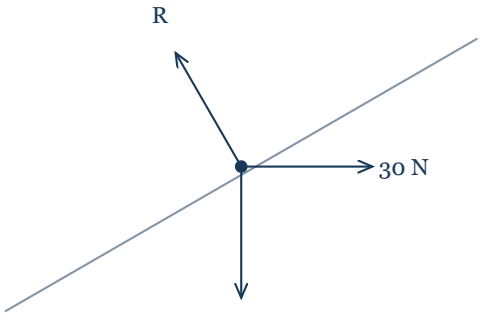
$$mg\sin 30^\circ = ma$$

$$a = g\sin 30^\circ = 4.9\,\text{m s}^{-2}$$

down the slope.

Exercise 5B · Question 6

A 5 kg particle on a smooth 30° slope is pushed by a horizontal 30 N force towards the rising slope. Find its acceleration.



Force directions · schematic

Answer

The component of the applied force up the slope is $30\cos 30°$.

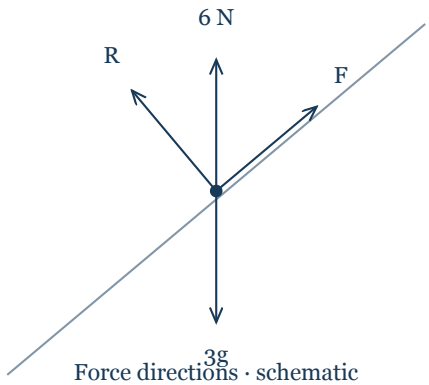
$30\cos 30° - 5g\sin 30° = 5a$

$a = 0.296\ \text{m s}^{-2}$

up the slope.

Exercise 5B · Question 7

A 3 kg particle on a rough 40° slope has a vertically upward force of 6 N applied. It moves at constant velocity. Find the frictional force.



Force directions · schematic

Answer

The resultant of weight and the applied vertical force is $(3g - 6)$ N downwards. Friction acts up the slope.

$F = (3g - 6)\sin 40°$

$F = 15.0\ \text{N}.$

Exercise 5B · Question 8

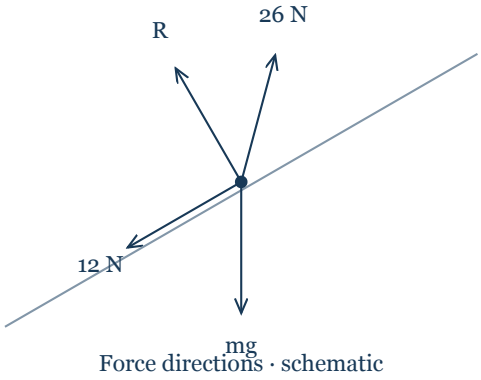
A particle of mass m moves up a rough slope with $\tan\alpha = 1/\sqrt{3}$. A 26 N force acts 45° above the plane. Friction is 12 N down the plane and the acceleration is $1\,\mathrm{m\,s^{-2}}$ up the plane. Find m to 3 significant figures. (3 marks)

Answer

$\alpha = 30^\circ$. Resolving up the plane:

$$26\cos 45^\circ - 12 - mg\sin 30^\circ = m.$$

$$m = \frac{13\sqrt{2} - 12}{1 + g/2} = 1.08\,\mathrm{kg}.$$



Force directions · schematic

Friction adjusts to the situation

1. No horizontal force on a stationary object on a horizontal surface: $F = 0$.
2. Pull T on a stationary object: friction balances T, until the limiting value is reached.
3. At limiting equilibrium, or sliding in the model used here:

   $F = \mu R$.

Friction opposes motion, or the tendency to move.

Below the limit:

$0 \le F \le \mu R$.

Do not automatically use $F = \mu R$ for every stationary particle.

Friction

Solution 21

Coefficient of friction

$F_{\max} = \mu R$.

μ is dimensionless and describes the pair of surfaces.

R is the normal reaction, not automatically the weight.

Rougher surfaces generally have a larger coefficient. A larger normal reaction increases the limiting friction.

On an inclined plane, resolve perpendicular to the plane first. Applied forces can change R.

μ need not be less than 1.

Illustrative static coefficients from the source

Steel on steel: 0.74; aluminium on steel: 0.61.

Copper on steel: 0.53; rubber on concrete: 1.0.

Wood on wood: 0.25–0.5; glass on glass: 0.94.

Waxed wood on wet snow: 0.14.

Ice on ice: 0.10; Teflon on Teflon: 0.04.

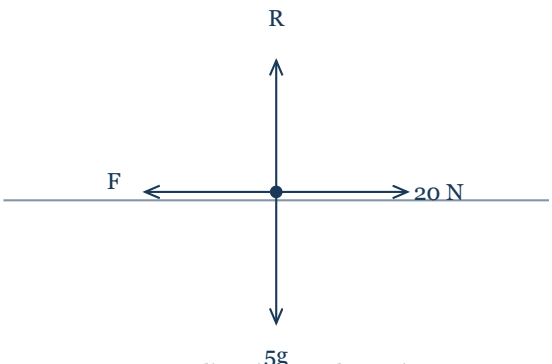
Synovial joints: 0.01.

No value was supplied for waxed wood on dry snow or lubricated metal on metal.

Example

Solution 22

A 5 kg particle is pulled along a rough horizontal surface by a horizontal force of 20 N. The coefficient of friction is 0.2. Find its acceleration.



Force directions · schematic

Answer

$R = 5g = 49\,\text{N},$

$F = \mu R = 0.2(49) = 9.8\,\text{N}.$

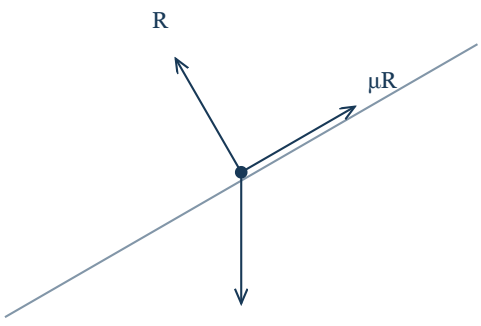
$20 - 9.8 = 5a$

$a = 2.04\,\text{m s}^{-2}.$

Inclined Plane with Friction Example

Solution 23

A 2 kg particle slides down a rough 30° plane with acceleration 1 m s^{-2}. Find μ.



Force directions · schematic

Answer

$R = 2g\cos 30°$.

Down the plane:

$2g\sin 30° - \mu R = 2(1)$.

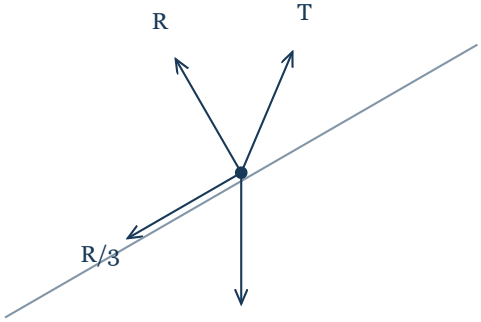
$$\mu = \frac{g-2}{2g\cos 30°} = 0.460.$$

Test Your Understanding

Solution 24

Edexcel M1 June 2013 · Question 3

A box of mass 2 kg rests on a rough plane inclined at 30°. A light inextensible rope pulls at angle α above the plane, where $\tan\alpha = 3/4$. The coefficient of friction is $1/3$. The box is on the point of moving up the plane. Model it as a particle and find the tension. (8 marks)



Force directions · schematic

Answer

$\sin\alpha = 3/5,\ \cos\alpha = 4/5$.

Perpendicular to the plane:

$$R + \frac{3}{5}T = 2g\cos 30°.$$

Friction is $R/3$ down the plane:

$$\frac{4}{5}T - \frac{R}{3} = 2g\sin 30°.$$

Substitute for R:

$$T = g\left(1 + \frac{1}{\sqrt{3}}\right) = 15.5\,\text{N}.$$

Exercise 5C · Question 3

A 0.5 kg particle slides down a rough 15° slope with acceleration $0.25\,\text{m s}^{-2}$. Find μ to 3 significant figures. (3 marks)

3

$R = 0.5g\cos 15^\circ,$

$0.5g\sin 15^\circ - \mu R = 0.5(0.25).$

$\mu = \dfrac{g\sin 15^\circ - 0.25}{g\cos 15^\circ} = 0.242.$

Question 4

A 2 kg particle slides down a rough 20° plane. A force P acts up the plane. Given $\mu = 0.3$ and acceleration $0.2\,\text{m s}^{-2}$ down the plane, find P. (4 marks)

4

$R = 2g\cos 20^\circ,$

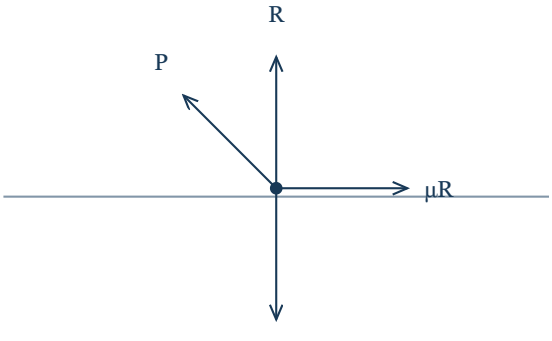
$2g\sin 20^\circ - P - 0.3R = 2(0.2).$

$P = 2g\sin 20^\circ - 0.6g\cos 20^\circ - 0.4$

$= 0.778\,\text{N}.$

Exercise 5C · Question 6

A sled of mass 10 kg is pulled across rough horizontal ground by a force P at 45° above the horizontal towards the left. Given $\mu = 0.1$ and acceleration $0.3\,\text{m s}^{-2}$ left, find P.



Force directions · schematic

Answer

$R + P\sin 45^\circ = 10g.$

Resolving left:

$P\cos 45^\circ - 0.1R = 10(0.3).$

$P(\cos 45^\circ + 0.1\sin 45^\circ) = 3 + g.$

$P = 16.5\,\text{N}.$

Exercise 5C · Question 7

A train travelling at $20\,\mathrm{m\,s^{-1}}$ stops in $30\,\mathrm{s}$ after its wheels lock.

(a) Model the train as a particle and assume friction is the only resistance. Find μ.

a

$$a = \frac{0 - 20}{30} = -\frac{2}{3}\,\mathrm{m\,s^{-2}}.$$

$$R = mg, \quad -\mu mg = -\frac{2}{3}m$$

$$\Rightarrow \mu = \frac{2}{3g} = 0.0680.$$

(b) How would allowing for air resistance change the inferred coefficient? (2 marks)

b

It would decrease. Air resistance also contributes to the observed deceleration, so less friction is required.

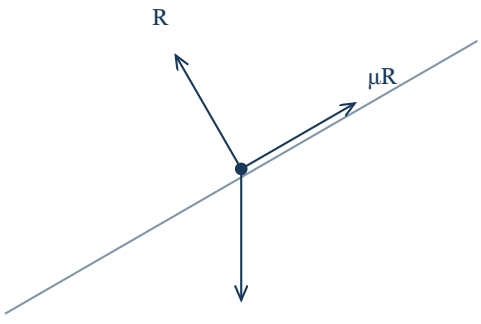
With air resistance A:

$$\mu mg + A = \frac{2}{3}m$$

$$\Rightarrow \mu = \frac{2}{3g} - \frac{A}{mg}.$$

Additional Question

A particle is on the point of sliding down a rough plane inclined at θ. Prove that $\mu = \tan\theta$. Can μ exceed 1?



Force directions · schematic

Answer

$R = mg\cos\theta$.

Limiting equilibrium along the plane:

$mg\sin\theta = \mu R = \mu mg\cos\theta$.

$$\boxed{\mu = \tan\theta}.$$

Yes: when $\theta > 45°$, $\tan\theta > 1$.